

# BRAF fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes BRAF gene fusions from the msk\_impact\_50k\_2026 study, covering 179 fusion events. 151/179 fusions (84.4%) were in-frame. 163/179 fusions (91.1%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ( $p=0.0133676$ , statistically significant at  $\alpha=0.05$ ). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 84.4% in-frame and 91.1% domain retention.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 179 analyzed fusion events, identifying 67 distinct fusion partner genes. The most frequent partner was KIAA1549, observed in 43 events.

### Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions retained the domain,  $p=0.0133676$  (statistically significant at  $\alpha=0.05$ ). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00699301.

### Domain disruption

Domain disruption was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions disrupted the domain,  $p=0.041705$  (statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.963504.

### Cutpoint detection

Cutpoint detection scanned 178 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 380 aa (exon 8; permutation-corrected  $p=0.001998$ , statistically significant at  $\alpha=0.05$ ). This is -78 aa from the nearest configured domain boundary at 458 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=163$ ) and "not\_retained" ( $n=15$ ). For Frame\_status, retained: 142/152 (93.4%, 95% CI 88.3%-96.4%); not\_retained: 9/11 (81.8%, 95% CI 52.3%-94.9%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 52.0736 vs 36.4,  $p=0.11965$  (not statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score produced results for this run (67 row(s) in composite\_evidence\_ranking); see the accompanying table.

## Exon retention

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## Joint partner

Joint partner reported: BRAF has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| KIAA1549     | 43          | 0.24022346368715083  | 0.21553360374650618    | 0.001614663595055688    | 0.8847211802950737       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.33366411984501465 | 1    |
| CUL1         | 5           | 0.027932960893854747 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28565588873695824 | 2    |
| FAM131B      | 3           | 0.01675977653631285  | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28230393342969573 | 3    |
| SND1         | 17          | 0.09497206703910614  | 0.21553360374650618    | 0.001614663595055688    | 0.8317520556609741       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.28214333215548626 | 4    |
| AKAP9        | 2           | 0.0111731843575419   | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2806279557760644  | 5    |
| CARMIL1      | 2           | 0.0111731843575419   | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2806279557760644  | 6    |
| EXOC4        | 2           | 0.0111731843575419   | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2806279557760644  | 7    |
| MAD1L1       | 2           | 0.0111731843575419   | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2806279557760644  | 8    |
| ABCC1        | 1           | 0.00558659217877095  | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331  | 9    |

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score    | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|--------------------|------|
| CAPZA2       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331 | 10   |
| CCDC6        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331 | 11   |
| CMAS         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331 | 12   |
| ERC1         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331 | 13   |
| KLHL12       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331 | 14   |
| LRBA         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331 | 15   |
| MINDY4       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331 | 16   |
| OSBPL7       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331 | 17   |
| PHTF2        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331 | 18   |
| PRIM2        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331 | 19   |

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| PRKAR1B      | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331  | 20   |
| PTPRZ1       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331  | 21   |
| RGS3         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331  | 22   |
| SNX8         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331  | 23   |
| TRA2B        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.989247311827957        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2789519781224331  | 24   |
| TRIM24       | 8           | 0.0446927374301676  | 0.21553360374650618    | 0.001614663595055688    | 0.8951612903225806       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.27657091847204573 | 25   |
| NRF1         | 2           | 0.0111731843575419  | 0.21553360374650618    | 0.001614663595055688    | 0.9247311827956989       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2709505364212257  | 26   |
| AGAP3        | 6           | 0.0335195530726257  | 0.21553360374650618    | 0.001614663595055688    | 0.8637992831541219       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.26851466208951436 | 27   |
| GTF2IRD1     | 3           | 0.01675977653631285 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.26294909472001826 | 28   |
| ATF7         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 29   |

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| CLEC2L       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 30   |
| DBI          | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 31   |
| DOCK4        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 32   |
| GIPC2        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 33   |
| GTF2I        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 34   |
| KCND2        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 35   |
| NFIA         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 36   |
| OSBPL9       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 37   |
| PAK2         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 38   |
| PRKAR2B      | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 39   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| RRBP1        | 1           | 0.00558659217877095  | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 40   |
| SETBP1       | 1           | 0.00558659217877095  | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 41   |
| SLC45A3      | 1           | 0.00558659217877095  | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 42   |
| SPOCK1       | 1           | 0.00558659217877095  | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 43   |
| ZC3HAV1      | 1           | 0.00558659217877095  | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 44   |
| ZNF207       | 1           | 0.00558659217877095  | 0.21553360374650618    | 0.001614663595055688    | 0.8602150537634409       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25959713941275575 | 45   |
| CDK5RAP2     | 4           | 0.0223463687150838   | 0.21553360374650618    | 0.001614663595055688    | 0.803763440860215        | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2561573304381657  | 46   |
| CLIP1        | 2           | 0.0111731843575419   | 0.21553360374650618    | 0.001614663595055688    | 0.7096774193548387       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2386924719050967  | 47   |
| AGK          | 14          | 0.0782122905027933   | 0.21553360374650618    | 0.001614663595055688    | 0.41167434715821816      | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.21410374291917902 | 48   |
| MKRN1        | 10          | 0.055865921787709494 | 0.21553360374650618    | 0.001614663595055688    | 0.4354838709677419       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.21097126087608245 | 49   |

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| TMPRSS2      | 4           | 0.0223463687150838  | 0.21553360374650618    | 0.001614663595055688    | 0.4408602150537635       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.20172184656719797 | 50   |
| ABCC2        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.4301075268817204       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.19508101038049763 | 51   |
| ITGB5        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.4301075268817204       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.19508101038049763 | 52   |
| MKLN1        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.4301075268817204       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.19508101038049763 | 53   |
| NEO1         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.4301075268817204       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.19508101038049763 | 54   |
| PWWP2A       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.4301075268817204       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.19508101038049763 | 55   |
| TBXAS1       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.4301075268817204       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.19508101038049763 | 56   |
| UBE2H        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.4301075268817204       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.19508101038049763 | 57   |
| WEE2         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.4301075268817204       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.19508101038049763 | 58   |
| PJA2         | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.3763440860215054       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.1870164942514654  | 59   |

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| SVOPL        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.3763440860215054       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.1870164942514654  | 60   |
| EPB41        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.3655913978494624       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.18540359102565895 | 61   |
| PLPP1        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.3655913978494624       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.18540359102565895 | 62   |
| RBM33        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.3655913978494624       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.18540359102565895 | 63   |
| RBMS3        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.3655913978494624       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.18540359102565895 | 64   |
| SCRIB        | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.3655913978494624       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.18540359102565895 | 65   |
| TRIM33       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.3655913978494624       | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.18540359102565895 | 66   |
| LUC7L2       | 1           | 0.00558659217877095 | 0.21553360374650618    | 0.001614663595055688    | 0.0                      | 0.7468257003897063         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.1305648813482396  | 67   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio          | p_value             | neg_log10_p_value  |
|----------|------------------------|------------------|------------------------|------------------|---------------------|---------------------|--------------------|
| 238      | 1                      | 162              | 0                      | 15               | None                | 1.0                 | -0.0               |
| 287      | 1                      | 162              | 1                      | 14               | 0.08641975308641975 | 0.16187392877547135 | 0.7908230925727642 |
| 303      | 2                      | 161              | 1                      | 14               | 0.17391304347826086 | 0.23330512802756184 | 0.6320757153442778 |

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio          | p_value               | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|---------------------|-----------------------|---------------------|
| 327      | 19                     | 144              | 6                      | 9                | 0.19791666666666666 | 0.00883450422715351   | 2.053817817119616   |
| 380      | 19                     | 144              | 9                      | 6                | 0.08796296296296297 | 4.496509404813937e-05 | 4.347124493887456   |
| 381      | 114                    | 49               | 9                      | 6                | 1.5510204081632653  | 0.5596655980047007    | 0.2520713878510084  |
| 393      | 140                    | 23               | 12                     | 3                | 1.5217391304347827  | 0.46313499579777323   | 0.33429240123968906 |
| 400      | 141                    | 22               | 12                     | 3                | 1.6022727272727273  | 0.44662922460047316   | 0.3500528628421283  |
| 438      | 141                    | 22               | 14                     | 1                | 0.4577922077922078  | 0.6965277300948016    | 0.15706158880581256 |
| 439      | 163                    | 0                | 14                     | 1                | None                | 0.08426966292134831   | 1.0743287432532127  |

## domain\_disruption tables

*frame\_domain\_contingency\_table*

|     |    |
|-----|----|
| 142 | 22 |
| 9   | 5  |

*domain\_disruption\_descriptives*

| Domain_key    | Domain_name          | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|---------------|----------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| ras_binding   | RAS-binding domain   | 178                     | 14                   | 0               | 164              | 0.0                                             |
| cysteine_rich | Cysteine-rich domain | 178                     | 14                   | 1               | 163              | 0.005699893955461294                            |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|     |    |
|-----|----|
| 142 | 21 |
| 9   | 6  |

*domain\_retention\_descriptives*

| Domain_key | Domain_name           | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|------------|-----------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| kinase     | Protein kinase domain | 178                     | 163                  | 0               | 15               | 0.0                                             |

## exon\_retention tables

*Exon\_retention*

| Exon | Retained_event_count | Total_event_count | Retained_fraction  |
|------|----------------------|-------------------|--------------------|
| 11   | 163                  | 179               | 0.9106145251396648 |

## frequency tables

### *Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| ABCC1        | 1           |
| ABCC2        | 1           |
| AGAP3        | 6           |
| AGK          | 14          |
| AKAP9        | 2           |
| ATF7         | 1           |
| CAPZA2       | 1           |
| CARMIL1      | 2           |
| CCDC6        | 1           |
| CDK5RAP2     | 4           |
| CLEC2L       | 1           |
| CLIP1        | 2           |
| CMAS         | 1           |
| CUL1         | 5           |
| DBI          | 1           |
| DOCK4        | 1           |
| EPB41        | 1           |
| ERC1         | 1           |
| EXOC4        | 2           |
| FAM131B      | 3           |
| GIPC2        | 1           |
| GTF2I        | 1           |
| GTF2IRD1     | 3           |
| ITGB5        | 1           |
| KCND2        | 1           |
| KIAA1549     | 43          |
| KLHL12       | 1           |
| LRBA         | 1           |
| LUC7L2       | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| MAD1L1       | 2           |
| MINDY4       | 1           |
| MKLN1        | 1           |
| MKRN1        | 10          |
| NEO1         | 1           |
| NFIA         | 1           |
| NRF1         | 2           |
| OSBPL7       | 1           |
| OSBPL9       | 1           |
| PAK2         | 1           |
| PHTF2        | 1           |
| PJA2         | 1           |
| PLPP1        | 1           |
| PRIM2        | 1           |
| PRKAR1B      | 1           |
| PRKAR2B      | 1           |
| PTPRZ1       | 1           |
| PWWP2A       | 1           |
| RBM33        | 1           |
| RBMS3        | 1           |
| RGS3         | 1           |
| RRBP1        | 1           |
| SCRIB        | 1           |
| SETBP1       | 1           |
| SLC45A3      | 1           |
| SND1         | 17          |
| SNX8         | 1           |
| SPOCK1       | 1           |
| SVOPL        | 1           |
| TBXAS1       | 1           |
| TMPRSS2      | 4           |
| TRA2B        | 1           |
| TRIM24       | 8           |
| TRIM33       | 1           |
| UBE2H        | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| WEE2         | 1           |
| ZC3HAV1      | 1           |
| ZNF207       | 1           |

## Per-event results (results.tsv)

| event_id                        | sampl_e_id            | pa_tie_nt_id | fusion_name | part_ner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                              | source_site2_effect_on_frame |
|---------------------------------|-----------------------|--------------|-------------|---------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>1458-T01-I<br>M6-1 | P-0021<br>458-T01-IM6 |              | ABCC1-BRAF  | ABCC1         | in-frame     | threonine   | 140493152          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | ABCC1 (NM_004996) - BRAF (NM_004333) rearrangement: t(7;16)(q34,p13.11)(chr7:g.140493152::chr16:g.16116105)   Protein Fusion: in frame {ABCC1:BRAF} | NA                           |
| EVT-P-003<br>4621-T01-I<br>M6-2 | P-0034<br>621-T01-IM6 |              | ABCC2-BRAF  | ABCC2         | unknown      | threonine   | 140495500          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | ABCC2 (NM_000392) - BRAF (NM_004333) rearrangement: t(7;10)(q34;24.2)(chr7:g.140495500::chr10:g.101559038)   Protein Fusion: mid-exon {ABCC2:BRAF}  | NA                           |
| EVT-P-005<br>2582-T01-I<br>M6-3 | P-0052<br>582-T01-IM6 |              | AGAP3-BRAF  | AGAP3         | unknown      | threonine   | 140485658          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1637:AGAP3_c.1177+1690:BRAFinv   Protein Fusion: mid-exon {AGAP3:BRAF}                               | NA                           |
| EVT-P-005<br>0574-T02-I<br>M6-4 | P-0050<br>574-T02-IM6 |              | AGAP3-BRAF  | AGAP3         | in-frame     | threonine   | 140493208          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1222-2059:AGAP3_c.1140+900:BRAFinv   Protein Fusion: in frame {AGAP3:BRAF}                           | NA                           |

| event_id                        | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                    | source_site_2_effect_on_frame |
|---------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>9799-T01-I<br>M6-5 | P-0039<br>799-T01-IM6 |            | AGAP3-BRAF  | AGAP3        | in-frame     | three_prime | 140482317          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1221+416:AGAP3_c.1314+504:BRAFInv   Protein Fusion: in frame {AGAP3:BRAF}  | NA                            |
| EVT-P-005<br>0931-T01-I<br>M6-6 | P-0050<br>931-T01-IM6 |            | AGK-BRAF    | AGK          | in-frame     | three_prime | 140499251          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-14294:AGK_c.980+911:BRAFInv   Protein Fusion: in frame {AGK:BRAF}        | NA                            |
| EVT-P-004<br>2835-T03-I<br>M6-7 | P-0042<br>835-T03-IM6 |            | AGK-BRAF    | AGK          | in-frame     | three_prime | 140497995          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv   Protein Fusion: in frame {AGK:BRAF}       | NA                            |
| EVT-P-004<br>2835-T02-I<br>M6-8 | P-0042<br>835-T02-IM6 |            | AGK-BRAF    | AGK          | in-frame     | three_prime | 140497995          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv   Protein Fusion: in frame {AGK:BRAF}       | NA                            |
| EVT-P-005<br>0635-T01-I<br>M6-9 | P-0050<br>635-T01-IM6 |            | AKAP9-BRAF  | AKAP9        | in-frame     | three_prime | 140487501          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.6613-1489:AKAP9_c.1141-117:BRAFInv   Protein Fusion: in frame {AKAP9:BRAF} | NA                            |

| event_id                     | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                             | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>0237-T03-IM6-10 | P-0030<br>237-T03-IM6 |            | AKAP9-BRAF  | AKAP9        | in-frame     | three_prime | 140492357          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.3318+1378:AKAP9_c.1140+1751:BRAF   Protein Fusion: in frame {AKAP9:BRAF}                            | NA                           |
| EVT-P-002<br>3821-T01-IM6-12 | P-0023<br>821-T01-IM6 |            | ATF7-BRAF   | ATF7         | in-frame     | three_prime | 140483233          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | ATF7 (NM_006856) - BRAF (NM_004333) Rearrangement : t(7;12)(q34;q13.13)(chr7:g.140483233::chr12:g.53913299)   Protein Fusion: in frame {ATF7:BRAF} | NA                           |
| EVT-P-000<br>7861-T01-IM5-13 | P-0007<br>861-T01-IM5 |            | BRAF-AGAP3  | AGAP3        | in-frame     | three_prime | 140490697          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.1141-3313:BRAF   Protein fusion: in frame (AGAP3-BRAF)                      | NA                           |
| EVT-P-002<br>1519-T02-IM6-14 | P-0021<br>519-T02-IM6 |            | BRAF-AGAP3  | AGAP3        | in-frame     | three_prime | 140493112          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement : c.1326+142:AGAP3_c.1140+996:BRAF   Protein Fusion: in frame {AGAP3:BRAF}                      | NA                           |
| EVT-P-000<br>9757-T01-IM5-15 | P-0009<br>757-T01-IM5 |            | BRAF-AGAP3  | AGAP3        | in-frame     | three_prime | 140493871          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1222-2154:AGAP3_c.1140+237:BRAF   Protein fusion: in frame (AGAP3-BRAF)                      | NA                           |

| event_id                         | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                         | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                        | source_site2_effect_on_frame |
|----------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------------------------------|------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>6886-T01-I<br>M6-16 | P-0036<br>886-T01-IM6 |            | BRAF-AGK    | AGK          | in-frame     | three_prime | 140494309          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+10859:AGK_c.981-42:BRAFInv   Protein Fusion: in frame {AGK:BRAF}                                      | NA                           |
| EVT-P-002<br>5561-T03-I<br>M6-17 | P-0025<br>561-T03-IM6 |            | BRAF-AGK    | AGK          | in-frame     | three_prime | 140498719          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 8-18): c.980+1288:BRAF_c.101+772:AGKInv   Protein Fusion: in frame {AGK:BRAF}    | NA                           |
| EVT-P-002<br>5561-T01-I<br>M6-18 | P-0025<br>561-T01-IM6 |            | BRAF-AGK    | AGK          | in-frame     | five_prime  | 140498874          | 7               | 327                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | BRAF (NM_004333) - AGK (NM_018238) Rearrangement: c.980+1288:BRAF_c.101+772:AGKInv   Protein Fusion: in frame {BRAF:AGK}                                      | NA                           |
| EVT-P-002<br>4227-T01-I<br>M6-19 | P-0024<br>227-T01-IM6 |            | BRAF-AGK    | AGK          | in-frame     | three_prime | 140495070          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+16796:AGK_c.981-803:BRAFInv   Protein Fusion: in frame {AGK:BRAF}                                     | NA                           |
| EVT-P-000<br>4477-T01-I<br>M5-20 | P-0004<br>477-T01-IM5 |            | BRAF-AGK    | AGK          | in-frame     | three_prime | 140495405          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076:AGK_c.981-1138:BRAFInv   Protein fusion: in frame (AGK-BRAF) | NA                           |
| EVT-P-002<br>3378-T01-I<br>M6-21 | P-0023<br>378-T01-IM6 |            | BRAF-AGK    | AGK          | in-frame     | five_prime  | 140498362          | 7               | 327                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | BRAF (NM_004333) - AGK (NM_018238) rearrangement: c.980+1800:BRAF_c.101+14574:AGKInv   Protein Fusion: in frame {BRAF:AGK}                                    | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                          | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0001832-T02-IM6-22 | P-0001832-T02-IM6 |            | BRAF-AGK    | AGK          | in-frame     | three_prime | 140494489          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exon 1 fused in-frame to BRAF exons 8-18): c.102-572:AGK_c.981-222:BRAFinv   Protein fusion: in frame (AGK-BRAF) | NA                           |
| EVT-P-0000913-T02-IM5-23 | P-0000913-T02-IM5 |            | BRAF-AGK    | AGK          | in-frame     | three_prime | 140496386          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAFinv   Protein fusion: in frame (AGK-BRAF)        | NA                           |
| EVT-P-0029531-T01-IM6-24 | P-0029531-T01-IM6 |            | BRAF-AGK    | AGK          | unknown      | three_prime | 140500233          | 7               | 303                         | False                  | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 7-18): c.101+4572:AGK_c.909:BRAFinv   Protein Fusion: mid-exon {AGK:BRAF}          | NA                           |
| EVT-P-0030746-T01-IM6-25 | P-0030746-T01-IM6 |            | BRAF-AGK    | AGK          | in-frame     | three_prime | 140497533          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BRAFinv   Protein Fusion: in frame {AGK:BRAF}                                              | NA                           |
| EVT-P-0010945-T01-IM5-26 | P-0010945-T01-IM5 |            | BRAF-AGK    | AGK          | in-frame     | three_prime | 140497610          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFinv   Protein fusion: in frame (AGK-BRAF)                                     | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name           | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                         | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                                                              | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-----------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------------------------------|------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-004<br>9035-T01-I<br>M6-85 | P-0049<br>035-T0<br>1-IM6 |            | BRAF-<br>CLEC<br>2L   | CLEC<br>2L   | in-frame     | five_prime  | 140485<br>799      | 9               | 393                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | BRAF (NM_004333) - CLEC2L (NM_001080511) rearrangement: c.1177+1549:BRAF_c.191-5932:CLEC2 Linv   Protein Fusion: in frame {BRAF:CLEC2L}                                                             | NA                           |
| EVT-P-000<br>4795-T01-I<br>M5-87 | P-0004<br>795-T0<br>1-IM5 |            | BRAF-<br>CUL1         | CUL1         | in-frame     | three_prime | 140493<br>811      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | CUL1(NM_003592) - BRAF (NM_004333) Rearrangement :c.790-427: CUL1_c.1140+297:BRAFinv   Protein fusion: in frame (CUL1-BRAF)                                                                         | NA                           |
| EVT-P-002<br>2257-T01-I<br>M6-88 | P-0022<br>257-T0<br>1-IM6 |            | BRAF-<br>EPB4<br>1    | EPB41        | in-frame     | three_prime | 140482<br>687      | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | EPB41 (NM_001166005) - BRAF (NM_004333) fusion (EPB41 exons 1-5 fused in-frame with BRAF exons 11-18): t(1;7)(p35.4;q34)(chr1:g.29424227::chr7:g.140482687)   Protein Fusion: in frame {EPB41:BRAF} | NA                           |
| EVT-P-000<br>1988-T01-I<br>M3-89 | P-0001<br>988-T0<br>1-IM3 |            | BRAF-<br>FAM1<br>31B  | FAM131B      | in-frame     | three_prime | 140489<br>629      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | FAM131B (NM_001031690) - BRAF (NM_004333) Deletion : c.175-103:FAM131B_c.1141-2245:BRAFdel   Protein fusion: in frame (FAM131B-BRAF)                                                                | NA                           |
| EVT-P-001<br>0494-T01-I<br>M5-90 | P-0010<br>494-T0<br>1-IM5 |            | BRAF-<br>GIPC2        | GIPC2        | in-frame     | three_prime | 140487<br>276      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | GIPC2 (NM_017655) -BRAF (NM_004333) Rearrangement : t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276)   Protein fusion: in frame (GIPC2-BRAF)                                                    | NA                           |
| EVT-P-005<br>4046-T01-I<br>M6-93 | P-0054<br>046-T0<br>1-IM6 |            | BRAF-<br>GTF2I<br>RD1 | GTF2IRD1     | in-frame     | five_prime  | 140484<br>706      | 9               | 393                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-1749:BRAF_c.2320+10446:GTF2IRD1inv   Protein Fusion: in frame {BRAF:GTF2IRD1}                                                         | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name           | partner_gene     | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                         | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                 | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------------|------------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------------------------------|------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>0828-T03-I<br>M6-94  | P-0000<br>828-T0<br>3-IM6 |            | BRAF-<br>GTF2I<br>RD1 | GTF<br>2IR<br>D1 | in-frame     | five_prime  | 140483<br>554      | 9               | 393                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-597: BRAF_c.2320+14181:GTF2IRD1inv   Protein Fusion: in frame {BRAF:GTF2IRD1}            | NA                            |
| EVT-P-001<br>5847-T02-I<br>M6-95  | P-0015<br>847-T0<br>2-IM6 |            | BRAF-<br>ITGB5        | ITGB5            | out-of-frame | five_prime  | 140497<br>019      | 7               | 327                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | BRAF (NM_004333) - ITGB5 (NM_002213) Rearrangement : t(3,7)(q21.2;q34)(chr3:g.124579522::chr7:g.140497019)   Protein Fusion: out of frame {BRAF:ITGB5} | NA                            |
| EVT-P-001<br>4377-T01-I<br>M6-99  | P-0014<br>377-T0<br>1-IM6 |            | BRAF-<br>NFIA         | NFIA             | in-frame     | three_prime | 140486<br>673      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | NFIA (NM_001145512) - BRAF (NM_004333) rearrangement: t(1;7)(p31.3;q34)(chr1:g.61836194::chr7:g.140486673)   Protein fusion: in frame (NFIA-BRAF)      | NA                            |
| EVT-P-001<br>0398-T01-I<br>M5-100 | P-0010<br>398-T0<br>1-IM5 |            | BRAF-<br>OSBPL9       | OSBPL9           | in-frame     | three_prime | 140484<br>912      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | OSBPL9 (NM_148909) - BRAF (NM_004333) Rearrangement : t(1;7)(p32.3;q34)(chr1:g.52228623::chr7:g.140484912)   Protein fusion: in frame (OSBPL9-BRAF)    | NA                            |
| EVT-P-000<br>3502-T01-I<br>M5-101 | P-0003<br>502-T0<br>1-IM5 |            | BRAF-<br>PJA2         | PJA2             | in-frame     | five_prime  | 140482<br>379      | 10              | 438                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | NA   Protein fusion: in frame (BRAF-PJA2)                                                                                                              | NA                            |
| EVT-P-003<br>6870-T01-I<br>M6-102 | P-0036<br>870-T0<br>1-IM6 |            | BRAF-<br>PRIM2        | PRIM2            | in-frame     | three_prime | 140493<br>232      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | PRIM2 (NM_000947) - BRAF (NM_004333) Rearrangement : t(6;7)(p11.1;q34)(chr6:g.57410815::chr7:g.140493232)   Protein Fusion: in frame {PRIM2:BRAF}      | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name          | partner_gene    | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                          | lost_domains                                              | disrupted_domains | source_annotation_text                                                                                                                                                                                     | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|----------------------|-----------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------------------------------------|-----------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-000<br>9246-T01-I<br>M5-103 | P-0009<br>246-T0<br>1-IM5 |            | BRAF-<br>RBM3<br>3   | RB<br>M33       | in-frame     | three_prime | 140482<br>661      | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                               | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | RBM33 (NM_053043) -<br>BRAF (NM_004333)<br>rearrangement: c.171+526:<br>RBM33_c.1314+160:BRAF<br>inv   Protein fusion: in frame<br>(RBM33-BRAF)                                                            | NA                           |
| EVT-P-001<br>6313-T01-I<br>M6-104 | P-0016<br>313-T0<br>1-IM6 |            | BRAF-<br>SLC4<br>5A3 | SLC<br>45A<br>3 | unknown      | three_prime | 140486<br>608      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                               | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | SLC45A3 (NM_033102) -<br>BRAF (NM_004333) fusion<br>(SLC45A3 exons 1 fused to<br>BRAF exons 10-18): t(1;7)(q<br>32.1;q34)(chr1:g.20563845<br>2::chr7:g.140486608)  <br>Transcript Fusion<br>{SLC45A3:BRAF} | NA                           |
| EVT-P-006<br>4764-T01-I<br>M7-105 | P-0064<br>764-T0<br>1-IM7 |            | BRAF-<br>SND1        | SND<br>1        | in-frame     | five_prime  | 140493<br>289      | 8               | 380                         | True                   | lost          | 0.0                      | False               | RAS-bi<br>nding<br>domain;<br>Cysteine-<br>rich<br>domain | Protein<br>kinase<br>domain                               |                   | BRAF (NM_004333) - SND1<br>(NM_014390) fusion: c.1140<br>+819:BRAF_c.1039-3950:S<br>ND1inv   Protein Fusion: in<br>frame {BRAF:SND1}                                                                       | NA                           |
| EVT-P-002<br>9981-T01-I<br>M6-106 | P-0029<br>981-T0<br>1-IM6 |            | BRAF-<br>SPOC<br>K1  | SPO<br>CK1      | in-frame     | three_prime | 140483<br>862      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                               | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | SPOCK1 (NM_004598) -<br>BRAF (NM_004333) fusion:<br>t(5;7)(q31.2;q34)(chr5:g.136<br>401840::chr7:g.140483862)  <br>Protein Fusion: in frame<br>{SPOCK1:BRAF}                                               | NA                           |
| EVT-P-006<br>4223-T01-I<br>M7-107 | P-0064<br>223-T0<br>1-IM7 |            | BRAF-<br>TBXA<br>S1  | TBX<br>AS1      | out-of-frame | five_prime  | 140494<br>512      | 7               | 327                         | True                   | lost          | 0.0                      | False               | RAS-bi<br>nding<br>domain;<br>Cysteine-<br>rich<br>domain | Protein<br>kinase<br>domain                               |                   | BRAF (NM_004333) -<br>TBXAS1 (NM_001061)<br>rearrangement: c.981-245:B<br>RAF_c.336+2985:TBXAS1<br>inv   Protein Fusion: out of<br>frame {BRAF:TBXAS1}                                                     | NA                           |
| EVT-P-002<br>1469-T01-I<br>M6-108 | P-0021<br>469-T0<br>1-IM6 |            | BRAF-<br>TRIM3<br>3  | TRI<br>M33      | in-frame     | three_prime | 140481<br>644      | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                               | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | TRIM33 (NM_015906) -<br>BRAF(NM_004333)<br>rearrangement: t(1;7)(p13.2;<br>q34)(chr1:g.114963985::chr<br>7:g.140481644)   Protein<br>Fusion: in frame<br>{TRIM33:BRAF}                                     | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                         | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                                                      | source_site_2_effect_on_frame |
|-------------------------------|-----------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------------------------------|------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-004<br>2835-T03-IM6-109 | P-0042<br>835-T03-IM6 |            | BRAF-WEE2    | WEE2         | unknown      | five_prime  | 140498912          | 7               | 327                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | BRAF (NM_004333) - WEE2 (NM_001105558) rearrangement: c.980+1250 :BRAF_c.1305:WEE2inv   Protein Fusion: mid-exon {BRAF:WEE2}                                                                | NA                            |
| EVT-P-001<br>5985-T03-IM6-111 | P-0015<br>985-T03-IM6 |            | CAPZA2-BRAF  | CAPZA2       | in-frame     | three_prime | 140491425          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | CAPZA2 (NM_006136) - BRAF (NM_004333) fusion: c.40-938:CAPZA2_c.1140+2683:BRAFinv   Protein Fusion: in frame {CAPZA2:BRAF}                                                                  | NA                            |
| EVT-P-003<br>3764-T04-IM6-112 | P-0033<br>764-T04-IM6 |            | CARMIL1-BRAF | CARMIL1      | in-frame     | three_prime | 140491509          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | LRRC16A (NM_017640) - BRAF (NM_004333) fusion: t(6;7)(p22.2;q34)(chr6:g.25554753::chr7:g.140491509)   Protein Fusion: in frame {LRRC16A:BRAF}                                               | NA                            |
| EVT-P-003<br>0796-T01-IM6-113 | P-0030<br>796-T01-IM6 |            | CARMIL1-BRAF | CARMIL1      | in-frame     | three_prime | 140491645          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | LRRC16A (NM_017640) - BRAF (NM_004333) fusion (LRRC16A exons 1-29 fused to BRAF exons 9-18): t(6;7)(p22.2;q34)(chr6:g.25580104::chr7:g.140491645)   Protein Fusion: in frame {LRRC16A:BRAF} | NA                            |
| EVT-P-000<br>5834-T01-IM5-114 | P-0005<br>834-T01-IM5 |            | CCDC6-BRAF   | CCDC6        | in-frame     | three_prime | 140488401          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | CCDC6 (NM_005436) -BRAF (NM_004333) Rearrangement : t(10;7)(chr10q21.2;q34)(chr10:g.61627617::chr7:g.140488401)   Protein fusion: in frame (CCDC6-BRAF)                                     | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name           | partner_gene      | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                  | lost_domains                                              | disrupted_domains | source_annotation_text                                                                                                                                               | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------------|-------------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------|-----------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>8194-T01-I<br>M6-115 | P-0038<br>194-T0<br>1-IM6 |            | CDK5<br>RAP2-<br>BRAF | CDK<br>5RAP<br>P2 | in-frame     | three_prime | 140484<br>954      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                       | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | CDK5RAP2 (NM_018249) -<br>BRAF (NM_004333)<br>rearrangement: t(7;9)(q34;q33.2)(chr7:g.140484954::chr9:g.123187156)   Protein<br>Fusion: in frame<br>{CDK5RAP2:BRAF}  | NA                            |
| EVT-P-000<br>4203-T01-I<br>M5-116 | P-0004<br>203-T0<br>1-IM5 |            | CDK5<br>RAP2-<br>BRAF | CDK<br>5RAP<br>P2 | in-frame     | three_prime | 140491<br>585      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                       | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | CDK5RAP2 (NM_018249) -<br>BRAF (NM_004333)<br>rearrangement: t(7;9)(7q34;q33.2)(chr7:140491585::chr9:123251876)   Protein<br>fusion: in frame<br>(CDK5RAP2-BRAF)     | NA                            |
| EVT-P-000<br>5461-T02-I<br>M5-117 | P-0005<br>461-T0<br>2-IM5 |            | CDK5<br>RAP2-<br>BRAF | CDK<br>5RAP<br>P2 | in-frame     | three_prime | 140481<br>684      | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                       | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | CDK5RAP2 (NM_018249) -<br>BRAF (NM_004333)<br>rearrangement : t(7;9)(7q34;q33.2)(chr7:140481684::chr9:123255452)   Protein<br>fusion: in frame<br>(CDK5RAP2-BRAF)    | NA                            |
| EVT-P-001<br>4969-T01-I<br>M6-118 | P-0014<br>969-T0<br>1-IM6 |            | CDK5<br>RAP2-<br>BRAF | CDK<br>5RAP<br>P2 | in-frame     | five_prime  | 140493<br>700      | 8               | 380                         | True                   | lost          | 0.0                      | False               | RAS-binding<br>domain;<br>Cysteine-rich<br>domain | Protein<br>kinase<br>domain                               |                   | BRAF (NM_004333) -<br>CDK5RAP2 (NM_018249)<br>Rearrangement : t(7;9)(q34;q33.3)(chr7:g.140493700::chr9:g.123269531)   Protein<br>fusion: in frame<br>(BRAF-CDK5RAP2) | NA                            |
| EVT-P-004<br>0461-T01-I<br>M6-119 | P-0040<br>461-T0<br>1-IM6 |            | CLIP1<br>-BRAF        | CLIP1             | in-frame     | three_prime | 140488<br>218      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                       | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | CLIP1 (NM_001247997) -<br>BRAF (NM_004333) fusion:<br>t(7;12)(q34;q24.31)(chr7:g.140488218::chr12:g.122833264)   Protein Fusion: in<br>frame {CLIP1:BRAF}            | NA                            |

| event_id                  | sampl_e_id        | pa_tie_nt_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                           | source_site_2_effect_on_frame |
|---------------------------|-------------------|--------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0007266-T02-IM6-120 | P-0007266-T02-IM6 |              | CLIP1-BRAF  | CLIP1        | in-frame     | three_prime | 140495673          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140495673::chr12:g.122767026)   Protein Fusion: in frame {CLIP1:BRAF} | NA                            |
| EVT-P-0024995-T01-IM6-121 | P-0024995-T01-IM6 |              | CMAS-BRAF   | CMAS         | unknown      | three_prime | 140488046          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | BRAF (NM_004333) rearrangement: t(7;12)(q34;p12.1)(chr7:g.140488046::chr12:g.22218755)   Transcript Fusion {CMAS:BRAF}                           | NA                            |
| EVT-P-0039648-T01-IM6-122 | P-0039648-T01-IM6 |              | CUL1-BRAF   | CUL1         | in-frame     | three_prime | 140492893          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215: BRAFInv   Protein Fusion: in frame {CUL1:BRAF}                          | NA                            |
| EVT-P-0039648-T03-IM6-123 | P-0039648-T03-IM6 |              | CUL1-BRAF   | CUL1         | in-frame     | three_prime | 140492893          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215: BRAFInv   Protein Fusion: in frame {CUL1:BRAF}                          | NA                            |
| EVT-P-0064088-T01-IM7-124 | P-0064088-T01-IM7 |              | CUL1-BRAF   | CUL1         | in-frame     | three_prime | 140489451          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+1956:CUL1_c.1141-2067: BRAFInv   Protein Fusion: in frame {CUL1:BRAF}                          | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                    | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-006<br>8173-T01-I<br>M7-125 | P-0068<br>173-T0<br>1-IM7 |            | CUL1-BRAF   | CUL1         | in-frame     | three_prime | 140491990          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+191:CUL1_c.1140+2118:BRAFInv   Protein Fusion: in frame {CUL1:BRAF}                     | NA                           |
| EVT-P-005<br>0196-T01-I<br>M6-126 | P-0050<br>196-T0<br>1-IM6 |            | DBI-BRAF    | DBI          | in-frame     | three_prime | 140486284          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | DBI (NM_001178017) - BRAF (NM_004333) fusion: t(2;7)(q14.2;q34)(chr2:g.120128894::chr7:g.140486284)   Protein Fusion: in frame {DBI:BRAF} | NA                           |
| EVT-P-006<br>4588-T01-I<br>M7-127 | P-0064<br>588-T0<br>1-IM7 |            | DOCK4-BRAF  | DOCK4        | out-of-frame | three_prime | 140483872          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | DOCK4 (NM_014705) - BRAF (NM_004333) fusion: c.4650+298:DOCK4_c.1178-915:BRAFdup   Protein Fusion: out of frame {DOCK4:BRAF}              | NA                           |
| EVT-P-001<br>4833-T02-I<br>M6-128 | P-0014<br>833-T0<br>2-IM6 |            | ERC1-BRAF   | ERC1         | in-frame     | three_prime | 140490668          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | ERC1 (NM_178040) - BRAF (NM_004333) fusion: t(7;12)(q34;p13.33)(chr7:g.140490668::chr12:g.1246645)   Protein Fusion: in frame {ERC1:BRAF} | NA                           |
| EVT-P-003<br>6937-T01-I<br>M6-129 | P-0036<br>937-T0<br>1-IM6 |            | EXOC4-BRAF  | EXOC4        | out-of-frame | three_prime | 140490241          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | EXOC4 (NM_021807) - BRAF (NM_004333) fusion: c.1514+93469:EXOC4_c.1141-2857:BRAFInv   Protein Fusion: out of frame {EXOC4:BRAF}           | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name           | part_ne_gene     | fra_me_status    | tar_get_role             | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_truncated | retain_ed_domains               | lost_domains                                                              | disrupt_ed_domains | source_annotation_text                                                                                                                                      | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|-----------------------|------------------|------------------|--------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|---------------------|---------------------------------|---------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>4206-T01-I<br>M6-130 | P-0024<br>206-T0<br>1-IM6 |              | EXOC<br>4-BRA<br>F    | EXO<br>C4        | out-of-fr<br>ame | thr_ee<br>_pr<br>im<br>e | 140487<br>956         | 9                | 381                          | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                    | EXOC4 (NM_021807) -<br>BRAF (NM_004333)<br>rearrangement: c.87-8072:E<br>XOC4_c.1141-572:BRAFinv<br>  Protein Fusion: out of<br>frame {EXOC4:BRAF}          | NA                            |
| EVT-P-004<br>7918-T04-I<br>M7-131 | P-0047<br>918-T0<br>4-IM7 |              | FAM1<br>31B-B<br>RAF  | FAM<br>131<br>B  | in-fr<br>ame     | thr_ee<br>_pr<br>im<br>e | 140487<br>437         | 9                | 381                          | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                    | FAM131B (NM_001031690)<br>- BRAF (NM_004333)<br>fusion: c.175-20:FAM131B_<br>c.1141-53:BRAFdel  <br>Protein Fusion: in frame<br>{FAM131B:BRAF}              | NA                            |
| EVT-P-004<br>7918-T02-I<br>M6-132 | P-0047<br>918-T0<br>2-IM6 |              | FAM1<br>31B-B<br>RAF  | FAM<br>131<br>B  | in-fr<br>ame     | thr_ee<br>_pr<br>im<br>e | 140487<br>437         | 9                | 381                          | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                    | FAM131B (NM_001031690)<br>- BRAF (NM_004333)<br>fusion: c.175-20:FAM131B_<br>c.1141-53:BRAFdel  <br>Protein Fusion: in frame<br>{FAM131B:BRAF}              | NA                            |
| EVT-P-004<br>4845-T01-I<br>M6-134 | P-0044<br>845-T0<br>1-IM6 |              | GTF2I<br>-BRAF        | GTF<br>2I        | unk<br>now<br>n  | thr_ee<br>_pr<br>im<br>e | 140485<br>631         | 10               | 393                          | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                    | GTF2I (NM_032999) -<br>BRAF (NM_004333) fusion:<br>c.768:GTF2I_c.1177+1717:<br>BRAFinv   Protein Fusion:<br>mid-exon {GTF2I:BRAF}                           | NA                            |
| EVT-P-000<br>0828-T05-I<br>M6-135 | P-0000<br>828-T0<br>5-IM6 |              | GTF2I<br>RD1-B<br>RAF | GTF<br>2IR<br>D1 | in-fr<br>ame     | thr_ee<br>_pr<br>im<br>e | 140483<br>546         | 10               | 393                          | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                    | GTF2IRD1 (NM_005685) -<br>BRAF (NM_004333)<br>rearrangement: c.2320+141<br>47:GTF2IRD1_c.1178-589:<br>BRAFinv   Protein Fusion: in<br>frame {GTF2IRD1:BRAF} | NA                            |

| event_id                      | sample_id             | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                                                  | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-004<br>6745-T01-IM6-136 | P-0046<br>745-T01-IM6 |            | KCND2-BRAF    | KCN D2       | out-of-frame | three_prime | 140485867          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KCND2 (NM_012281) - BRAF (NM_004333) fusion: c.1115+91797:KCND2_c.1177+1481:BRAFInv   Protein Fusion: out of frame {KCND2:BRAF}                                                         | NA                           |
| EVT-P-005<br>6952-T01-IM6-138 | P-0056<br>952-T01-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140493821          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-141:KIAA1549_c.1140+287:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                                    | NA                           |
| EVT-P-004<br>4270-T02-IM6-139 | P-0044<br>270-T02-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140491645          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1560:KIAA1549_c.1140+2463:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                                  | NA                           |
| EVT-P-002<br>7592-T01-IM6-140 | P-0027<br>592-T01-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140492343          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5247+769:KIAA1549_c.1140+1765:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF} | NA                           |
| EVT-P-005<br>5099-T03-IM6-141 | P-0055<br>099-T03-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140490663          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+1357:KIAA1549_c.1141-3279:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                                  | NA                           |

| event_id                          | sample_id             | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                                          | source_site_2_effect_on_frame |
|-----------------------------------|-----------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-005<br>8584-T01-I<br>M7-142 | P-0058<br>584-T01-IM7 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140491493          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1663:KIAA1549_c.1140+2615:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                          | NA                            |
| EVT-P-006<br>6030-T01-I<br>M7-143 | P-0066<br>030-T01-IM7 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140482785          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1172:KIAA1549_c.1314+36:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                            | NA                            |
| EVT-P-003<br>5153-T01-I<br>M6-144 | P-0035<br>153-T01-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140491093          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF} | NA                            |
| EVT-P-003<br>5153-T03-I<br>M6-145 | P-0035<br>153-T03-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140491093          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046:KIAA1549_c.1140+3015:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                          | NA                            |
| EVT-P-004<br>9925-T01-I<br>M6-146 | P-0049<br>925-T01-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140489497          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+301:KIAA1549_c.1141-2113:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                           | NA                            |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_nam<br>e   | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_ro<br>le     | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains        | lost_<br>do<br>mai<br>ns                                                  | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                                               | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|-----------------------|--------------------------|------------------------------|-----------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|---------------------------------|---------------------------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-003<br>5844-T03-I<br>M7-147 | P-0035<br>844-T0<br>3-IM7 |                            | KIAA1<br>549-B<br>RAF | KIA<br>A15<br>49         | in-fr<br>ame                 | thr<br>ee_<br>pr<br>im<br>e | 140493<br>285              | 9                           | 381                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                           | KIAA1549 (NM_001164665)<br>- BRAF (NM_004333)<br>fusion: c.4033-1202:KIAA15<br>49_c.1140+823:BRAFdup  <br>Protein Fusion: in frame<br>{KIAA1549:BRAF}                                                | NA                                   |
| EVT-P-003<br>5844-T04-I<br>M7-148 | P-0035<br>844-T0<br>4-IM7 |                            | KIAA1<br>549-B<br>RAF | KIA<br>A15<br>49         | in-fr<br>ame                 | thr<br>ee_<br>pr<br>im<br>e | 140493<br>285              | 9                           | 381                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                           | KIAA1549 (NM_001164665)<br>- BRAF (NM_004333)<br>fusion: c.4033-1202:KIAA15<br>49_c.1140+823:BRAFdup  <br>Protein Fusion: in frame<br>{KIAA1549:BRAF}                                                | NA                                   |
| EVT-P-001<br>7693-T02-I<br>M6-149 | P-0017<br>693-T0<br>2-IM6 |                            | KIAA1<br>549-B<br>RAF | KIA<br>A15<br>49         | in-fr<br>ame                 | thr<br>ee_<br>pr<br>im<br>e | 140489<br>537              | 9                           | 381                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                           | KIAA1549 (NM_001164665)<br>- BRAF (NM_004333)<br>Fusion: c.4930-495:KIAA15<br>49_c.1141-2153:BRAFdup  <br>Protein Fusion: in frame<br>{KIAA1549:BRAF}                                                | NA                                   |
| EVT-P-006<br>2963-T01-I<br>M7-150 | P-0062<br>963-T0<br>1-IM7 |                            | KIAA1<br>549-B<br>RAF | KIA<br>A15<br>49         | in-fr<br>ame                 | thr<br>ee_<br>pr<br>im<br>e | 140482<br>198              | 11                          | 439                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                           | KIAA1549 (NM_001164665)<br>- BRAF (NM_004333)<br>fusion: c.4930-27:KIAA1549<br>_c.1314+623:BRAFdup  <br>Protein Fusion: in frame<br>{KIAA1549:BRAF}                                                  | NA                                   |
| EVT-P-001<br>2575-T01-I<br>M5-151 | P-0012<br>575-T0<br>1-IM5 |                            | KIAA1<br>549-B<br>RAF | KIA<br>A15<br>49         | in-fr<br>ame                 | thr<br>ee_<br>pr<br>im<br>e | 140490<br>295              | 9                           | 381                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                           | KIAA1549 (NM_001164665)<br>- BRAF (NM_004333) fusion<br>(KIAA1549 exons 1-15 with<br>BRAF exons 9-18): c.4929+<br>793:KIAA1549_c.1141-2911<br>:BRAFdup   Protein fusion:<br>in frame (KIAA1549-BRAF) | NA                                   |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion<br>_nam<br>e   | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_ro<br>le     | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains        | lost<br>_do<br>mai<br>ns                                                  | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                                                                                                              | source_site<br>2_effect_on<br>_frame |
|-----------------------------------|---------------------------|----------------------------|-----------------------|--------------------------|------------------------------|-----------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|---------------------------------|---------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-005<br>2411-T05-I<br>M7-152 | P-0052<br>411-T0<br>5-IM7 |                            | KIAA1<br>549-B<br>RAF | KIA<br>A15<br>49         | in-fr<br>ame                 | thr<br>ee_<br>pr<br>im<br>e | 140493<br>530              | 9                           | 381                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                           | KIAA1549 (NM_001164665)<br>- BRAF (NM_004333)<br>fusion: c.4930-2296:KIAA15<br>49_c.1140+578:BRAFdup  <br>Protein Fusion: in frame<br>{KIAA1549:BRAF}                                                                                                               | NA                                   |
| EVT-P-003<br>4117-T01-I<br>M6-153 | P-0034<br>117-T0<br>1-IM6 |                            | KIAA1<br>549-B<br>RAF | KIA<br>A15<br>49         | in-fr<br>ame                 | thr<br>ee_<br>pr<br>im<br>e | 140483<br>812              | 10                          | 393                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                           | KIAA1549 (NM_001164665)<br>- BRAF (NM_004333)<br>fusion: c.4346-57:KIAA1549<br>_c.1178-855:BRAFdup  <br>Protein Fusion: in frame<br>{KIAA1549:BRAF}                                                                                                                 | NA                                   |
| EVT-P-005<br>2411-T03-I<br>M6-154 | P-0052<br>411-T0<br>3-IM6 |                            | KIAA1<br>549-B<br>RAF | KIA<br>A15<br>49         | in-fr<br>ame                 | thr<br>ee_<br>pr<br>im<br>e | 140493<br>530              | 9                           | 381                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                           | KIAA1549 (NM_001164665)<br>- BRAF (NM_004333)<br>fusion: c.4930-2296:KIAA15<br>49_c.1140+578:BRAFdup  <br>Protein Fusion: in frame<br>{KIAA1549:BRAF}                                                                                                               | NA                                   |
| EVT-P-000<br>9097-T01-I<br>M5-155 | P-0009<br>097-T0<br>1-IM5 |                            | KIAA1<br>549-B<br>RAF | KIA<br>A15<br>49         | in-fr<br>ame                 | thr<br>ee_<br>pr<br>im<br>e | 140488<br>817              | 9                           | 381                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                           | KIAA1549 (NM_001164665)<br>- BRAF (NM_004333) fusion<br>(KIAA1549 exons 1-16<br>fused in frame with BRAF<br>exons 9-18, which include<br>the BRAF kinase domain): c<br>.5247+3837:KIAA1549_c.11<br>41-1433:BRAFdup   Protein<br>fusion: in frame<br>(KIAA1549-BRAF) | NA                                   |
| EVT-P-000<br>5121-T02-I<br>M5-156 | P-0005<br>121-T0<br>2-IM5 |                            | KIAA1<br>549-B<br>RAF | KIA<br>A15<br>49         | in-fr<br>ame                 | thr<br>ee_<br>pr<br>im<br>e | 140491<br>248              | 9                           | 381                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                           | KIAA1549 (NM_001164665)<br>- BRAF (NM_004333) fusion<br>(KIAA1549 exons 1-15 with<br>BRAF exons 9-18): c.4929+<br>2454:KIAA1549_c.1140+28<br>60:BRAFdup   Protein<br>fusion: in frame<br>(KIAA1549-BRAF)                                                            | NA                                   |

| event_id                          | sample_id             | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                                                | source_site2_effect_on_frame |
|-----------------------------------|-----------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-006<br>4872-T01-I<br>M7-157 | P-0064<br>872-T01-IM7 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140489277          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4032+4347:KIAA1549_c.1141-1893:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                                | NA                           |
| EVT-P-001<br>3777-T01-I<br>M5-158 | P-0013<br>777-T01-IM5 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140481674          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BRADFdup   Protein fusion: in frame (KIAA1549-BRAF)                                          | NA                           |
| EVT-P-004<br>0760-T01-I<br>M6-159 | P-0040<br>760-T01-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140481540          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627:KIAA1549_c.1315-47:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                                  | NA                           |
| EVT-P-001<br>4191-T01-I<br>M6-160 | P-0014<br>191-T01-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140492679          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion : c.5248-320: KIAA1549_c.1140+1429:BRAFdup   Protein fusion: in frame (KIAA1549-BRAF)                                               | NA                           |
| EVT-P-001<br>4381-T01-I<br>M6-161 | P-0014<br>381-T01-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140492888          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082:KIAA1549_c.1140+1220:BRAFdup   Protein fusion: in frame (KIAA1549-BRAF) | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                                                                                               | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>8611-T01-IM6-162 | P-0028<br>611-T01-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140482786          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                                 | NA                           |
| EVT-P-001<br>5540-T01-IM6-163 | P-0015<br>540-T01-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140489119          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.4930-69:KIAA1549_c.1141-1735:BRAFdup   Protein fusion: in frame (KIAA1549-BRAF) | NA                           |
| EVT-P-006<br>2070-T01-IM7-164 | P-0062<br>070-T01-IM7 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140490443          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1064:KIAA1549_c.1141-3059:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                                                                               | NA                           |
| EVT-P-002<br>2404-T01-IM6-165 | P-0022<br>404-T01-IM6 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140482177          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 11-18) : c.5248-2270:KIAA1549_c.1314+644:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                             | NA                           |
| EVT-P-000<br>4032-T01-IM5-166 | P-0004<br>032-T01-IM5 |            | KIAA1549-BRAF | KIAA1549     | in-frame     | three_prime | 140487892          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAFdup   Protein fusion: in frame (KIAA1549-BRAF)                                                       | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                                               | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-004<br>3372-T01-I<br>M6-167 | P-0043<br>372-T0<br>1-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | three_prime | 140487523          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2204:KIAA1549_c.1141-139:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                                | NA                           |
| EVT-P-002<br>8611-T02-I<br>M6-168 | P-0028<br>611-T0<br>2-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | three_prime | 140482786          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF} | NA                           |
| EVT-P-004<br>3874-T01-I<br>M6-169 | P-0043<br>874-T0<br>1-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | three_prime | 140490001          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-2907:KIAA1549_c.1141-2617:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                               | NA                           |
| EVT-P-006<br>9024-T04-I<br>M7-170 | P-0069<br>024-T0<br>4-IM7 |            | KIAA1549-B<br>RAF | KIAA1549     | unknown      | three_prime | 140492117          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+1991:BRAFdup   Protein Fusion: mid-exon {KIAA1549:BRAF}                                                    | NA                           |
| EVT-P-002<br>9361-T01-I<br>M6-171 | P-0029<br>361-T0<br>1-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | three_prime | 140489625          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                               | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                                                            | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>6557-T01-I<br>M6-172 | P-0016<br>557-T0<br>1-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | threonine_prime | 140488576          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955: KIAA1549_c.1141-1192:BR AFdup   Protein Fusion: in frame {KIAA1549:BRAF} | NA                            |
| EVT-P-003<br>2982-T01-I<br>M6-173 | P-0032<br>982-T0<br>1-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | threonine_prime | 140491506          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+2602:BR AFdup   Protein Fusion: in frame {KIAA1549:BRAF}           | NA                            |
| EVT-P-006<br>4310-T04-I<br>M7-174 | P-0064<br>310-T0<br>4-IM7 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | threonine_prime | 140493446          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560:KIAA1549_c.1140+662:BR AFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                                            | NA                            |
| EVT-P-002<br>1127-T01-I<br>M6-175 | P-0021<br>127-T0<br>1-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | threonine_prime | 140491741          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c.4032+3996:KIAA1549_c.1140+2367:BR AFdup   Protein Fusion: in frame {KIAA1549:BRAF}         | NA                            |
| EVT-P-001<br>7693-T01-I<br>M6-176 | P-0017<br>693-T0<br>1-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | threonine_prime | 140489537          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (c.1141-2153): c.4930-495: KIAA1549_c.1141-2153:BR AFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                             | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                                                  | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>0125-T01-I<br>M6-177 | P-0020<br>125-T0<br>1-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | unknown      | threepri    | 140492175          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c.5260:KIAA1549_c.1140+1933:BRAFdup   Protein Fusion: mid-exon {KIAA1549:BRAF}        | NA                            |
| EVT-P-004<br>5683-T02-I<br>M6-178 | P-0045<br>683-T0<br>2-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | unknown      | threepri    | 140492850          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5189:KIAA1549_c.1140+1258:BRAFdup   Protein Fusion: mid-exon {KIAA1549:BRAF}                                                       | NA                            |
| EVT-P-001<br>9127-T01-I<br>M6-179 | P-0019<br>127-T0<br>1-IM6 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | threepri    | 140490657          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c.5247+2110:KIAA1549_c.1141-3273:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}   | NA                            |
| EVT-P-006<br>6022-T01-I<br>M7-180 | P-0066<br>022-T0<br>1-IM7 |            | KIAA1549-B<br>RAF | KIAA1549     | in-frame     | threepri    | 140488035          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-4074:KIAA1549_c.1141-651:BRAFdup   Protein Fusion: in frame {KIAA1549:BRAF}                                                   | NA                            |
| EVT-P-003<br>6680-T01-I<br>M6-181 | P-0036<br>680-T0<br>1-IM6 |            | KLHL12-BRAF       | KLHL12       | in-frame     | threepri    | 140493984          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | KLHL12 (NM_021633) - BRAF (NM_004333) fusion (KLHL12 exons 1-4 fused to BRAF exons 9-18): t(1;7)(q32.1;q34)(chr1:g.202880970:chr7:g.140493984)   Protein Fusion: in frame {KLHL12:BRAF} | NA                            |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                         | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                             | source_site_2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------------------------------|------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0047833-T02-IM6-183 | P-0047833-T02-IM6 |            | LRBA-BRAF   | LRBA         | in-frame     | three_prime | 140493444          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | LRBA (NM_001199282) - BRAF (NM_004333) fusion: t(4;7)(q31.3;q34)(chr4:g.151502269::chr7:g.140493444)   Protein Fusion: in frame {LRBA:BRAF}                        | NA                            |
| EVT-P-0006412-T01-IM5-184 | P-0006412-T01-IM5 |            | LUC7L2-BRAF | LUC7L2       | unknown      | five_prime  | 140501109          | 6               | 287                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BR AFdel   Antisense fusion                                                  | NA                            |
| EVT-P-0026158-T04-IM6-185 | P-0026158-T04-IM6 |            | MAD1L1-BRAF | MAD1L1       | in-frame     | three_prime | 140491749          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | MAD1L1 (NM_001013837) - BRAF (NM_004333) fusion: c.1417-9863:MAD1L1_c.1140+2359:BRAFdup   Protein Fusion: in frame {MAD1L1:BRAF}                                   | NA                            |
| EVT-P-0026158-T01-IM6-186 | P-0026158-T01-IM6 |            | MAD1L1-BRAF | MAD1L1       | in-frame     | three_prime | 140491749          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | MAD1L1 (NM_001013837) - BRAF (NM_004333) Rearrangement : c.1417-9863_c.1140+2359dup   Protein Fusion: in frame {MAD1L1:BRAF}                                       | NA                            |
| EVT-P-0006601-T03-IM6-187 | P-0006601-T03-IM6 |            | MINDY4-BRAF | MINDY4       | in-frame     | three_prime | 140490696          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | FAM188B (NM_032222) - BRAF(NM_004333) Fusion (FAM188B exon5 fused with BRAF exon9) : c.664-366:FAM188B_c.1141-3312:BR AF   Protein Fusion: in frame {FAM188B:BRAF} | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                              | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>3220-T01-I<br>M6-188 | P-0023<br>220-T0<br>1-IM6 |            | MKLN<br>1-BRAF | MKLN1        | in-frame     | three_prime | 140496796          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | MKLN1 (NM_013255) - BRAF (NM_004333) rearrangement: c.98+4070:MKLN1_c.981-2529:BRAFinv   Protein Fusion: in frame {MKLN1:BRAF}      | NA                            |
| EVT-P-003<br>5393-T02-I<br>M6-189 | P-0035<br>393-T0<br>2-IM6 |            | MKRN<br>1-BRAF | MKRN1        | in-frame     | three_prime | 140488436          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.771+833:MKRN1_c.1141-1052:BRAFdup   Protein Fusion: in frame {MKRN1:BRAF}            | NA                            |
| EVT-P-000<br>9024-T01-I<br>M5-190 | P-0009<br>024-T0<br>1-IM5 |            | MKRN<br>1-BRAF | MKRN1        | in-frame     | three_prime | 140493485          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G:MKRN1_c.1140+623:BRAFdup   Protein fusion: in frame (MKRN1-BRAF)     | NA                            |
| EVT-P-000<br>8083-T03-I<br>M6-191 | P-0008<br>083-T0<br>3-IM6 |            | MKRN<br>1-BRAF | MKRN1        | out-of-frame | three_prime | 140484511          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.315-432:MKRN1_c.1178-1554:BRAFdup   Protein Fusion: out of frame {MKRN1:BRAF} | NA                            |
| EVT-P-000<br>1036-T01-I<br>M3-192 | P-0001<br>036-T0<br>1-IM3 |            | MKRN<br>1-BRAF | MKRN1        | in-frame     | three_prime | 140482250          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | NA   Protein fusion: in frame (MKRN1-BRAF)                                                                                          | NA                            |

| event_id                      | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                                                    | disrupted_domains | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|-----------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-004<br>7135-T01-IM6-193 | P-0047<br>135-T01-IM6 |            | MKRN1-BRAF  | MKRN1        | in-frame     | three_prime | 140492510          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain                        |                   | MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-972:MKRN1_c.1140+1598:BRAFdup   Protein Fusion: in frame {MKRN1:BRAF}                                             | NA                           |
| EVT-P-000<br>4373-T02-IM5-194 | P-0004<br>373-T02-IM5 |            | MKRN1-BRAF  | MKRN1        | unknown      | three_prime | 140434466          | 18              | 744                         | False                  | lost          | 0.0                      | False               |                       | Protein kinase domain; RAS-binding domain; Cysteine-rich domain |                   | MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351:MKRN1_c.2232:BRAFdup   Protein fusion: mid-exon (MKRN1-BRAF)                                          | NA                           |
| EVT-P-000<br>9024-T02-IM6-195 | P-0009<br>024-T02-IM6 |            | MKRN1-BRAF  | MKRN1        | in-frame     | three_prime | 140493485          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain                        |                   | MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-185:MKRN1_c.1140+623:BRAFdup   Protein Fusion: in frame {MKRN1:BRAF}                                              | NA                           |
| EVT-P-003<br>1896-T02-IM6-196 | P-0031<br>896-T02-IM6 |            | MKRN1-BRAF  | MKRN1        | in-frame     | three_prime | 140482581          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain                        |                   | MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-4 fused to BRAF exons 11-18): c.772-1018:MKRN1_c.1314+240:BRAFdup   Protein Fusion: in frame {MKRN1:BRAF} | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains                                              | disrupted_domains | source_annotation_text                                                                                                                                                                            | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------|-----------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>8083-T02-I<br>M6-197 | P-0008<br>083-T0<br>2-IM6 |            | MKRN<br>1-BRAF | MKRN1        | out-of-frame | three_prime | 140484<br>505      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | MKRN1 (NM_013446) -<br>BRAF (NM_004333) fusion<br>(MKRN1 exons 1-2 fused<br>with BRAF exons 10-18): c.<br>315-428:MKRN1_c.1178-15<br>48:BRAFdup   Protein<br>Fusion: out of frame<br>{MKRN1:BRAF} | NA                            |
| EVT-P-001<br>3536-T01-I<br>M5-198 | P-0013<br>536-T0<br>1-IM5 |            | MKRN<br>1-BRAF | MKRN1        | in-frame     | three_prime | 140483<br>618      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | MKRN1 (NM_013446) -<br>BRAF (NM_004333)<br>rearrangement: c.544+172:<br>MKRN1_c.1178-661:BRAFdup   Protein fusion: in frame<br>(MKRN1-BRAF)                                                       | NA                            |
| EVT-P-004<br>8036-T01-I<br>M6-199 | P-0048<br>036-T0<br>1-IM6 |            | NEO1-<br>BRAF  | NEO1         | in-frame     | three_prime | 140497<br>651      | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | NEO1 (NM_002499) -<br>BRAF (NM_004333) fusion:<br>t(7;15)(q34;q24.1)(chr7:g.14<br>0497651::chr15:g.73575682<br>)   Protein Fusion: in frame<br>{NEO1:BRAF}                                        | NA                            |
| EVT-P-004<br>3972-T03-I<br>M6-200 | P-0043<br>972-T0<br>3-IM6 |            | NRF1-<br>BRAF  | NRF1         | in-frame     | three_prime | 140487<br>712      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | NRF1 (NM_005011) - BRAF<br>(NM_004333) fusion: c.607-<br>7410:NRF1_c.1141-328:BR<br>AFInv   Protein Fusion: in<br>frame {NRF1:BRAF}                                                               | NA                            |
| EVT-P-006<br>5446-T01-I<br>M7-201 | P-0065<br>446-T0<br>1-IM7 |            | NRF1-<br>BRAF  | NRF1         | in-frame     | three_prime | 140483<br>810      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | NRF1 (NM_005011) - BRAF<br>(NM_004333) fusion: c.1349-<br>10893:NRF1_c.1178-853:BR<br>AFInv   Protein Fusion: in<br>frame {NRF1:BRAF}                                                             | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name      | partner_gene    | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains                                              | disrupted_domains | source_annotation_text                                                                                                                                            | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|------------------|-----------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------|-----------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-005<br>9002-T01-I<br>M7-202 | P-0059<br>002-T0<br>1-IM7 |            | OSBP<br>L7-BRAF  | OSB<br>PL7      | in-frame     | three_prime | 140491<br>530      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS<br>-binding<br>domain;<br>Cysteine<br>-rich<br>domain |                   | OSBPL7 (NM_145798) -<br>BRAF (NM_004333) fusion:<br>t(7;17)(q34;q21.32)(chr7:g.1<br>40491530::chr17:g.4589038<br>1)   Protein Fusion: in frame<br>{OSBPL7:BRAF}   | NA                            |
| EVT-P-005<br>4958-T02-I<br>M6-203 | P-0054<br>958-T0<br>2-IM6 |            | PAK2-<br>BRAF    | PAK<br>2        | in-frame     | three_prime | 140486<br>065      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS<br>-binding<br>domain;<br>Cysteine<br>-rich<br>domain |                   | PAK2 (NM_002577) - BRAF<br>(NM_004333)<br>rearrangement: t(3;7)(q29;q<br>34)(chr3:g.196530579::chr7:<br>g.140486065)   Protein<br>Fusion: in frame<br>{PAK2:BRAF} | NA                            |
| EVT-P-000<br>8789-T01-I<br>M5-205 | P-0008<br>789-T0<br>1-IM5 |            | PHTF<br>2-BRAF   | PHT<br>F2       | in-frame     | three_prime | 140489<br>236      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS<br>-binding<br>domain;<br>Cysteine<br>-rich<br>domain |                   | PHTF2 (NM_001127358) -<br>BRAF (NM_004333)<br>rearrangement: c.2224-367:<br>PHTF2_c.1141-1852:BRAF<br>nv   Protein fusion: in frame<br>{PHTF2-BRAF}               | NA                            |
| EVT-P-004<br>4819-T01-I<br>M6-206 | P-0044<br>819-T0<br>1-IM6 |            | PLPP<br>1-BRAF   | PLP<br>P1       | out-of-frame | three_prime | 140482<br>123      | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS<br>-binding<br>domain;<br>Cysteine<br>-rich<br>domain |                   | PPAP2A (NM_176895) -<br>BRAF (NM_004333) fusion:<br>t(5;7)(q11.2;q34)(chr5:g.547<br>89192::chr7:g.140482123)  <br>Protein Fusion: out of frame<br>{PPAP2A:BRAF}   | NA                            |
| EVT-P-000<br>9694-T01-I<br>M5-207 | P-0009<br>694-T0<br>1-IM5 |            | PRKAR1B-<br>BRAF | PRK<br>AR1<br>B | in-frame     | three_prime | 140491<br>855      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS<br>-binding<br>domain;<br>Cysteine<br>-rich<br>domain |                   | PRKAR1B<br>(NM_001164758) -BRAF<br>(NM_004333)<br>Rearrangement : c.892-255<br>1:PRKAR1B_c.1140+2253:<br>BRAFdup   Protein fusion: in<br>frame {PRKAR1B-BRAF}     | NA                            |

| event_id                          | sample_id             | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                  | source_site2_effect_on_frame |
|-----------------------------------|-----------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-000<br>8567-T01-I<br>M5-208 | P-0008<br>567-T01-IM5 |            | PRKAR2B-BRAF | PRKAR2B      | in-frame     | three_prime | 140486416          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063:PRKAR2B_c.1177+932:BRAFInv   Protein fusion: in frame (PRKAR2B-BRAF)                   | NA                           |
| EVT-P-004<br>1181-T01-I<br>M6-209 | P-0041<br>181-T01-IM6 |            | PTPRZ1-BRAF  | PTPRZ1       | in-frame     | three_prime | 140492428          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | PTPRZ1 (NM_002851) - BRAF (NM_004333) fusion: c.5367+457:PTPRZ1_c.1140+1680:BRAFInv   Protein Fusion: in frame {PTPRZ1:BRAF}                            | NA                           |
| EVT-P-001<br>2884-T01-I<br>M5-210 | P-0012<br>884-T01-IM5 |            | PWWP2A-BRAF  | PWWP2A       | in-frame     | three_prime | 140495243          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | PWWP2A (NM_001130864) - BRAF (NM_004333) Rearrangement : t(5;7)(q33.3;q34)(chr5:g.159539324::chr7:g.140495243)   Protein fusion: in frame (PWWP2A-BRAF) | NA                           |
| EVT-P-006<br>8260-T01-I<br>M7-211 | P-0068<br>260-T01-IM7 |            | RBMS3-BRAF   | RBMS3        | in-frame     | three_prime | 140482707          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | RBMS3 (NM_001003793) - BRAF (NM_004333) fusion: t(3;7)(p24.1;q34)(chr3:g.30030473::chr7:g.140482707)   Protein Fusion: in frame {RBMS3:BRAF}            | NA                           |
| EVT-P-006<br>4764-T01-I<br>M7-212 | P-0064<br>764-T01-IM7 |            | RGS3-BRAF    | RGS3         | out-of-frame | three_prime | 140493282          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | RGS3 (NM_144488) - BRAF (NM_004333) fusion: t(7;9)(q34;q32)(chr7:g.140493282::chr9:g.116291333)   Protein Fusion: out of frame {RGS3:BRAF}              | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                            | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-004<br>1308-T02-I<br>M6-213 | P-0041<br>308-T0<br>2-IM6 |            | RRBP1-BRAF  | RRBP1        | in-frame     | three_prime | 140485917          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | RRBP1 (NM_004587) - BRAF (NM_004333) fusion: t(7;20)(q34;p12.1)(chr7:g.140485917::chr20:g.17634546)   Protein Fusion: in frame {RRBP1:BRAF}       | NA                            |
| EVT-P-000<br>7392-T01-I<br>M5-214 | P-0007<br>392-T0<br>1-IM5 |            | SCRIB-BRAF  | SCRIB        | in-frame     | three_prime | 140482150          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7;8)(q34;q24.3)(chr7:g.140482150::chr8:g.144876675)   Protein fusion: in frame (SCRIB-BRAF) | NA                            |
| EVT-P-005<br>6639-T01-I<br>M6-215 | P-0056<br>639-T0<br>1-IM6 |            | SETBP1-BRAF | SETBP1       | in-frame     | three_prime | 140486594          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | SETBP1 (NM_015559) - BRAF (NM_004333) fusion: t(7;18)(q34;q12.3)(chr7:g.140486594::chr18:g.42638299)   Protein Fusion: in frame {SETBP1:BRAF}     | NA                            |
| EVT-P-004<br>8950-T01-I<br>M6-216 | P-0048<br>950-T0<br>1-IM6 |            | SND1-BRAF   | SND1         | in-frame     | three_prime | 140492033          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+30037:SND1_c.1140+2075:BRAFinv   Protein Fusion: in frame {SND1:BRAF}                          | NA                            |
| EVT-P-001<br>2395-T01-I<br>M5-217 | P-0012<br>395-T0<br>1-IM5 |            | SND1-BRAF   | SND1         | in-frame     | three_prime | 140493527          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1152+27174:SND1_c.1140+581:BRAFinv   Protein fusion: in frame (SND1-BRAF)                    | NA                            |

| event_id                          | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                         | source_site_2_effect_on_frame |
|-----------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>9619-T01-I<br>M5-218 | P-0009<br>619-T01-IM5 |            | SND1-BRAF   | SND1         | in-frame     | three_prime | 140482494          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1968+2394:SND1_c.1314+327:BRAFinv   Protein fusion: in frame (SND1-BRAF)  | NA                            |
| EVT-P-005<br>2299-T04-I<br>M6-219 | P-0052<br>299-T04-IM6 |            | SND1-BRAF   | SND1         | unknown      | three_prime | 140482935          | 10              | 400                         | False                  | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1038+2929:SND1_c.1200:BRAFinv   Protein Fusion: mid-exon {SND1:BRAF}             | NA                            |
| EVT-P-000<br>1687-T01-I<br>M3-220 | P-0001<br>687-T01-IM3 |            | SND1-BRAF   | SND1         | in-frame     | three_prime | 140490224          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | SND1 (NM_014390) - BRAF (NM_004333) inversion: c.1152+28381_c.1141-2840inv   Protein fusion: in frame (SND1-BRAF)              | NA                            |
| EVT-P-000<br>4103-T01-I<br>M5-221 | P-0004<br>103-T01-IM5 |            | SND1-BRAF   | SND1         | in-frame     | three_prime | 140481685          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1038+5799:SND1_c.1315-192:BRAFinv.   Protein fusion: in frame (SND1-BRAF) | NA                            |
| EVT-P-002<br>9727-T01-I<br>M6-222 | P-0029<br>727-T01-IM6 |            | SND1-BRAF   | SND1         | in-frame     | three_prime | 140482257          | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+12095:SND1_c.1314+564:BRAFinv   Protein Fusion: in frame {SND1:BRAF}        | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains                                              | disrupted_domains | source_annotation_text                                                                                                                                                 | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------|-----------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-006<br>8194-T01-I<br>M7-223 | P-0068<br>194-T0<br>1-IM7 |            | SND1-<br>BRAF | SND<br>1     | in-frame     | three_prime | 140489<br>175      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS<br>-binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+17686:SND1_c.1141-1791:BRAFInv   Protein Fusion: in frame {SND1:BRAF}                                               | NA                           |
| EVT-P-000<br>8140-T01-I<br>M5-224 | P-0008<br>140-T0<br>1-IM5 |            | SND1-<br>BRAF | SND<br>1     | in-frame     | three_prime | 140489<br>439      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS<br>-binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18) : c.1039-3849:SND1_c.1141-2055:BRAFInv   Protein fusion: in frame (SND1-BRAF)   | NA                           |
| EVT-P-003<br>4168-T01-I<br>M6-225 | P-0034<br>168-T0<br>1-IM6 |            | SND1-<br>BRAF | SND<br>1     | in-frame     | three_prime | 140489<br>421      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS<br>-binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused to BRAF exons 9-18): c.1152+6972:SND1_c.1141-2037:BRAFInv   Protein Fusion: in frame {SND1:BRAF}     | NA                           |
| EVT-P-002<br>5075-T01-I<br>M6-226 | P-0025<br>075-T0<br>1-IM6 |            | SND1-<br>BRAF | SND<br>1     | in-frame     | three_prime | 140490<br>546      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS<br>-binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1- 9 with BRAF exons 9-18) : c.1039-2471:SND1_c.1141-3162:BRAFInv   Protein Fusion: in frame {SND1:BRAF}        | NA                           |
| EVT-P-000<br>2071-T02-I<br>M5-227 | P-0002<br>071-T0<br>2-IM5 |            | SND1-<br>BRAF | SND<br>1     | in-frame     | three_prime | 140489<br>292      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS<br>-binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SND1_c.1141-1908:BRAFInv   Protein fusion: in frame (SND1-BRAF) | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                         | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                           | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|------------------------------------------|------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>2197-T01-I<br>M3-228 | P-0002<br>197-T0<br>1-IM3 |            | SND1-<br>BRAF  | SND<br>1     | in-frame     | five_prime  | 140488<br>303      | 8               | 380                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | c.1528-8436:SND1_c.1141-922:BRAFinv   Protein fusion: in frame (BRAF-SND1)                                                                                       | NA                            |
| EVT-P-000<br>7533-T01-I<br>M5-229 | P-0007<br>533-T0<br>1-IM5 |            | SND1-<br>BRAF  | SND<br>1     | in-frame     | three_prime | 140482<br>226      | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SND1_c.1314+595:BRAFinv   Protein fusion: in frame (SND1-BRAF) | NA                            |
| EVT-P-003<br>6567-T01-I<br>M6-230 | P-0036<br>567-T0<br>1-IM6 |            | SND1-<br>BRAF  | SND<br>1     | in-frame     | three_prime | 140493<br>552      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | SND1 (NM_014390) - BRAF (NM_004333) Fusion (SND1 exon10 fused with BRAF exon9) : c.1153-22013:SND1_c.1140+556:BRAFinv   Protein Fusion: in frame {SND1:BRAF}     | NA                            |
| EVT-P-003<br>4845-T01-I<br>M6-231 | P-0034<br>845-T0<br>1-IM6 |            | SND1-<br>BRAF  | SND<br>1     | in-frame     | three_prime | 140489<br>241      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1527+3658:SND1_c.1141-1857:BRAFinv   Protein Fusion: in frame {SND1:BRAF}                                   | NA                            |
| EVT-P-005<br>4364-T01-I<br>M6-232 | P-0054<br>364-T0<br>1-IM6 |            | SNX8-<br>BRAF  | SNX<br>8     | in-frame     | three_prime | 140488<br>730      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain                    | RAS-binding domain; Cysteine-rich domain |                   | SNX8 (NM_013321) - BRAF (NM_004333) fusion: c.1284+565:SNX8_c.1141-1346:BRAFdup   Protein Fusion: in frame {SNX8:BRAF}                                           | NA                            |
| EVT-P-002<br>6773-T01-I<br>M6-234 | P-0026<br>773-T0<br>1-IM6 |            | SVOP<br>L-BRAF | SVO<br>PL    | unknown      | five_prime  | 140482<br>097      | 10              | 438                         | True                   | lost          | 0.0                      | False               | RAS-binding domain; Cysteine-rich domain | Protein kinase domain                    |                   | BRAF (NM_004333) rearrangement: c.1315-604: BRAF_chr7:g.138277426del   Transcript Fusion {BRAF:SVOPL}                                                            | NA                            |

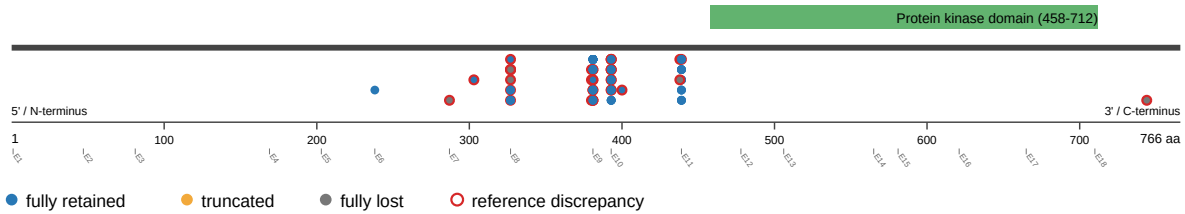
| event_id                          | sampl_e_id                | pa_tie_nt_id | fusion_name          | part_ne_ge      | frame_status     | target_role         | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                | lost_domains                                                              | disrupted_domains           | source_annotation_text                                                                                                                                                                | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|----------------------|-----------------|------------------|---------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------|---------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-005<br>1214-T01-I<br>M6-235 | P-0051<br>214-T0<br>1-IM6 |              | TMPR<br>SS2-B<br>RAF | TMP<br>RSS<br>2 | in-fr<br>ame     | thre<br>e_pr<br>ime | 140502<br>592      | 6               | 238                         | True                   | retai<br>ned  | 1.0                      | False               | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain                                         | Cystein<br>e-rich<br>domain | TMPRSS2<br>(NM_001135099) - BRAF<br>(NM_004333) fusion: t(7;21)<br>(q34;q22.3)(chr7:g.1405025<br>92::chr21:g.42867846)  <br>Protein Fusion: in frame<br>{TMPRSS2:BRAF}                | NA                            |
| EVT-P-001<br>8524-T01-I<br>M6-236 | P-0018<br>524-T0<br>1-IM6 |              | TMPR<br>SS2-B<br>RAF | TMP<br>RSS<br>2 | out-of-fr<br>ame | thre<br>e_pr<br>ime | 140490<br>337      | 9               | 381                         | True                   | retai<br>ned  | 1.0                      | False               | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                             | TMPRSS2<br>(NM_001135099) - BRAF<br>(NM_004333)<br>Rearrangement : t(21;7)(q2<br>2.3;q34)(chr21:g.42873801:<br>:chr7:g.140490337)  <br>Protein Fusion: out of frame<br>{TMPRSS2:BRAF} | NA                            |
| EVT-P-002<br>2778-T01-I<br>M6-237 | P-0022<br>778-T0<br>1-IM6 |              | TMPR<br>SS2-B<br>RAF | TMP<br>RSS<br>2 | in-fr<br>ame     | thre<br>e_pr<br>ime | 140483<br>147      | 10              | 393                         | True                   | retai<br>ned  | 1.0                      | False               | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                             | TMPRSS2<br>(NM_001135099) - BRAF<br>(NM_004333)<br>Rearrangement : t(7;21)(q3<br>4;q22.3)(chr7:g.140483147:<br>:chr21:g.42878796)  <br>Protein Fusion: in frame<br>{TMPRSS2:BRAF}     | NA                            |
| EVT-P-004<br>3836-T01-I<br>M6-239 | P-0043<br>836-T0<br>1-IM6 |              | TRA2<br>B-BR<br>AF   | TRA<br>2B       | in-fr<br>ame     | thre<br>e_pr<br>ime | 140491<br>978      | 9               | 381                         | True                   | retai<br>ned  | 1.0                      | False               | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                             | TRA2B (NM_004593) -<br>BRAF (NM_004333) fusion:<br>t(3;7)(q27.2;q34)(chr3:g.185<br>650761::chr7:g.140491978)<br>  Protein Fusion: in frame<br>{TRA2B:BRAF}                            | NA                            |
| EVT-P-005<br>4430-T02-I<br>M6-240 | P-0054<br>430-T0<br>2-IM6 |              | TRIM2<br>4-BRA<br>F  | TRI<br>M24      | in-fr<br>ame     | thre<br>e_pr<br>ime | 140487<br>875      | 9               | 381                         | True                   | retai<br>ned  | 1.0                      | False               | Protein<br>kinase<br>domai<br>n | RAS<br>-bin<br>ding<br>dom<br>ain;<br>Cyst<br>eine<br>-rich<br>dom<br>ain |                             | TRIM24 (NM_015905) -<br>BRAF (NM_004333) fusion:<br>c.2719-489:TRIM24_c.1141<br>-491:BRAFinv   Protein<br>Fusion: in frame<br>{TRIM24:BRAF}                                           | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains                                              | disrupted_domains | source_annotation_text                                                                                                                       | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------|-----------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-002<br>5699-T02-I<br>M6-241 | P-0025<br>699-T0<br>2-IM6 |            | TRIM2<br>4-BRAF | TRIM24       | in-frame     | three_prime | 140486<br>854      | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | TRIM24 (NM_015905) -<br>BRAF (NM_004333) fusion:<br>c.2014+674:TRIM24_c.117<br>7+494:BRAFInv   Protein<br>Fusion: in frame<br>{TRIM24:BRAF}  | NA                           |
| EVT-P-004<br>5111-T01-I<br>M6-242 | P-0045<br>111-T0<br>1-IM6 |            | TRIM2<br>4-BRAF | TRIM24       | in-frame     | three_prime | 140488<br>049      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | TRIM24 (NM_015905) -<br>BRAF (NM_004333) fusion:<br>c.2257-760:TRIM24_c.1141<br>-665:BRAFInv   Protein<br>Fusion: in frame<br>{TRIM24:BRAF}  | NA                           |
| EVT-P-006<br>6835-T01-I<br>M7-243 | P-0066<br>835-T0<br>1-IM7 |            | TRIM2<br>4-BRAF | TRIM24       | in-frame     | three_prime | 140490<br>589      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | TRIM24 (NM_015905) -<br>BRAF (NM_004333) fusion:<br>c.1879-502:TRIM24_c.1141<br>-3205:BRAFInv   Protein<br>Fusion: in frame<br>{TRIM24:BRAF} | NA                           |
| EVT-P-006<br>8451-T01-I<br>M7-244 | P-0068<br>451-T0<br>1-IM7 |            | TRIM2<br>4-BRAF | TRIM24       | in-frame     | three_prime | 140482<br>765      | 11              | 439                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | TRIM24 (NM_015905) -<br>BRAF (NM_004333) fusion:<br>c.484-860:TRIM24_c.1314+<br>56:BRAFInv   Protein<br>Fusion: in frame<br>{TRIM24:BRAF}    | NA                           |
| EVT-P-003<br>0988-T01-I<br>M6-245 | P-0030<br>988-T0<br>1-IM6 |            | TRIM2<br>4-BRAF | TRIM24       | unknown      | three_prime | 140491<br>002      | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | RAS-<br>binding<br>domain;<br>Cysteine-<br>rich<br>domain |                   | TRIM24 (NM_015905) -<br>BRAF (NM_004333) fusion:<br>c.2349:TRIM24_c.1140+31<br>06:BRAFInv   Protein<br>Fusion: mid-exon<br>{TRIM24:BRAF}     | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                  | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>2569-T01-I<br>M6-246 | P-0032<br>569-T0<br>1-IM6 |            | TRIM24-BRAF  | TRIM24       | in-frame     | three_prime | 140487936          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | TRIM24 (NM_015905) - BRAF (NM_004333) Rearrangement : c.1878+791:TRIM24_c.1141-552inv   Protein Fusion: in frame {TRIM24:BRAF}                          | NA                           |
| EVT-P-003<br>2569-T02-I<br>M6-247 | P-0032<br>569-T0<br>2-IM6 |            | TRIM24-BRAF  | TRIM24       | in-frame     | three_prime | 140487936          | 9               | 381                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1878+791:TRIM24_c.1141-552:BRAFinv   Protein Fusion: in frame {TRIM24:BRAF}                             | NA                           |
| EVT-P-001<br>9039-T01-I<br>M6-248 | P-0019<br>039-T0<br>1-IM6 |            | UBE2H-BRAF   | UBE2H        | out-of-frame | three_prime | 140499248          | 8               | 327                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | BRAF (NM_004333) - UBE2H (NM_003344) rearrangement: c.980+914: BRAF_c.299-4611:UBE2Hdup   Protein Fusion: out of frame {UBE2H:BRAF}                     | NA                           |
| EVT-P-002<br>7762-T01-I<br>M6-250 | P-0027<br>762-T0<br>1-IM6 |            | ZC3HAV1-BRAF | ZC3HAV1      | in-frame     | three_prime | 140485946          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | ZC3HAV1 (NM_020119) - BRAF (NM_004333) rearrangement: c.698-431:ZC3HAV1_c.1177+1402:BRAFdup   Protein Fusion: in frame {ZC3HAV1:BRAF}                   | NA                           |
| EVT-P-000<br>8920-T01-I<br>M5-251 | P-0008<br>920-T0<br>1-IM5 |            | ZNF207-BRAF  | ZNF207       | in-frame     | three_prime | 140485847          | 10              | 393                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | RAS-binding domain; Cysteine-rich domain |                   | ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr17:g.30687533)   Protein fusion: in frame (ZNF207-BRAF) | NA                           |

## domain retention outliers

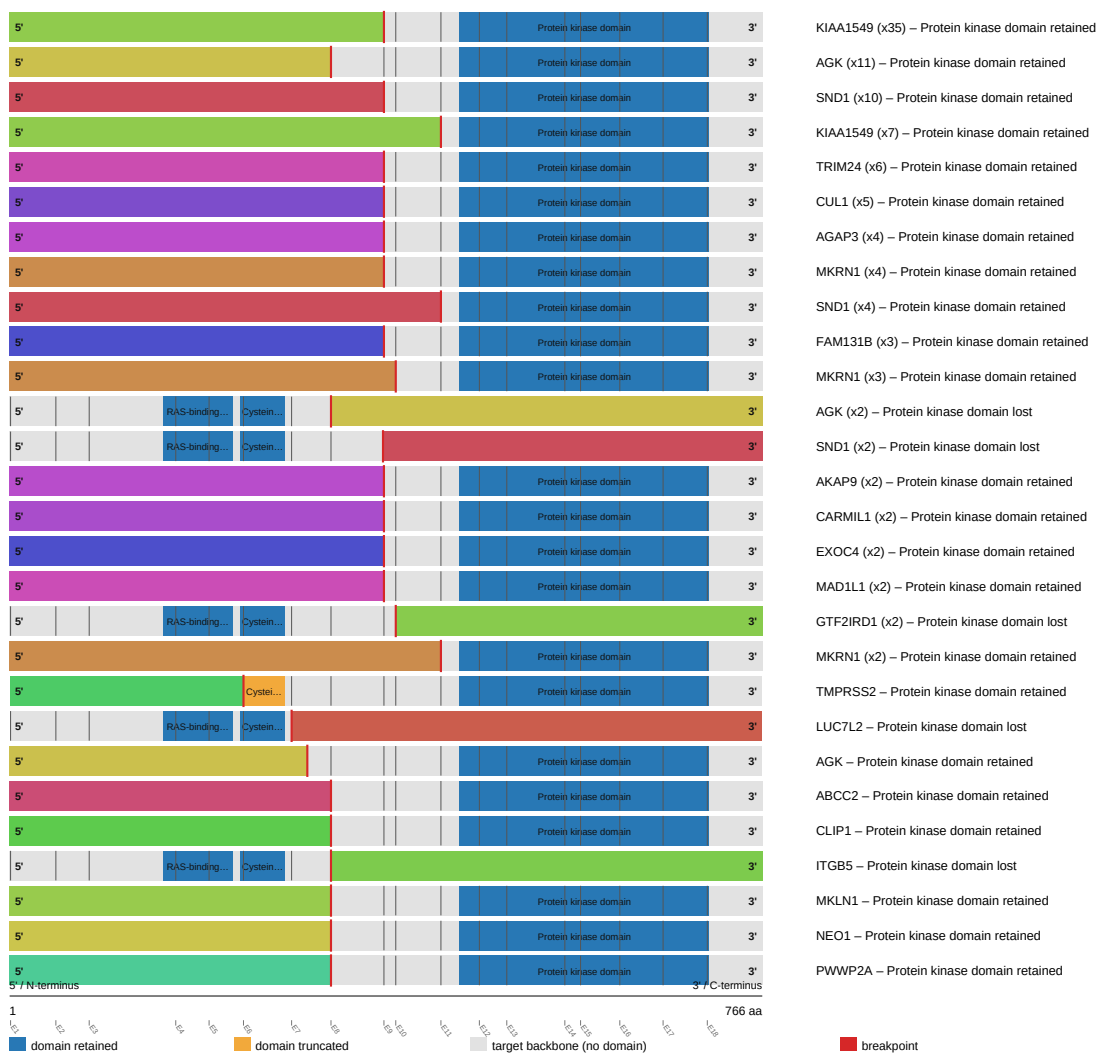
BRAF domain-retention track



## fusion schematic

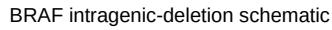
### BRAF fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (BRAF portion is domain-colored, exon ticks)

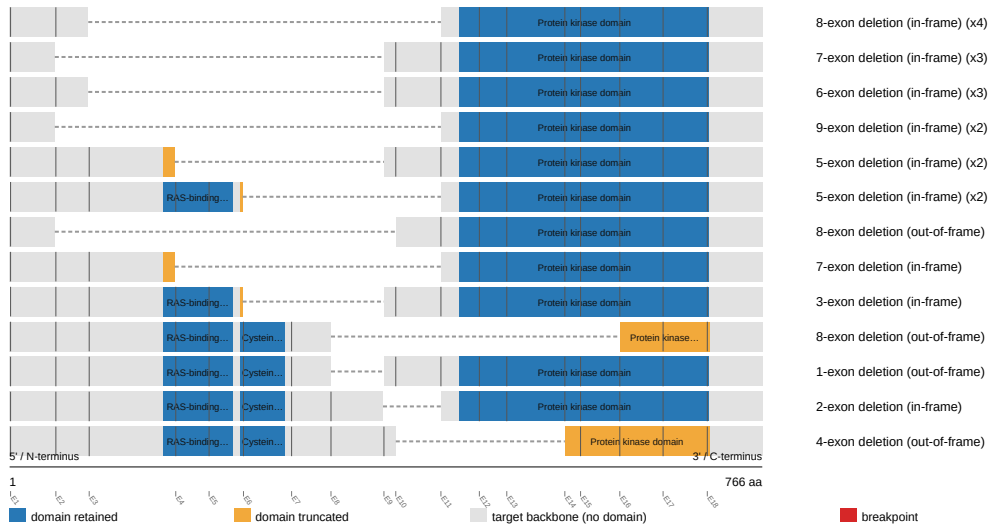


Showing the top 28 of 89 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

### intragenic deletion schematic



retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



## reference comparison

Reference vs msk\_impact\_50k\_2026

